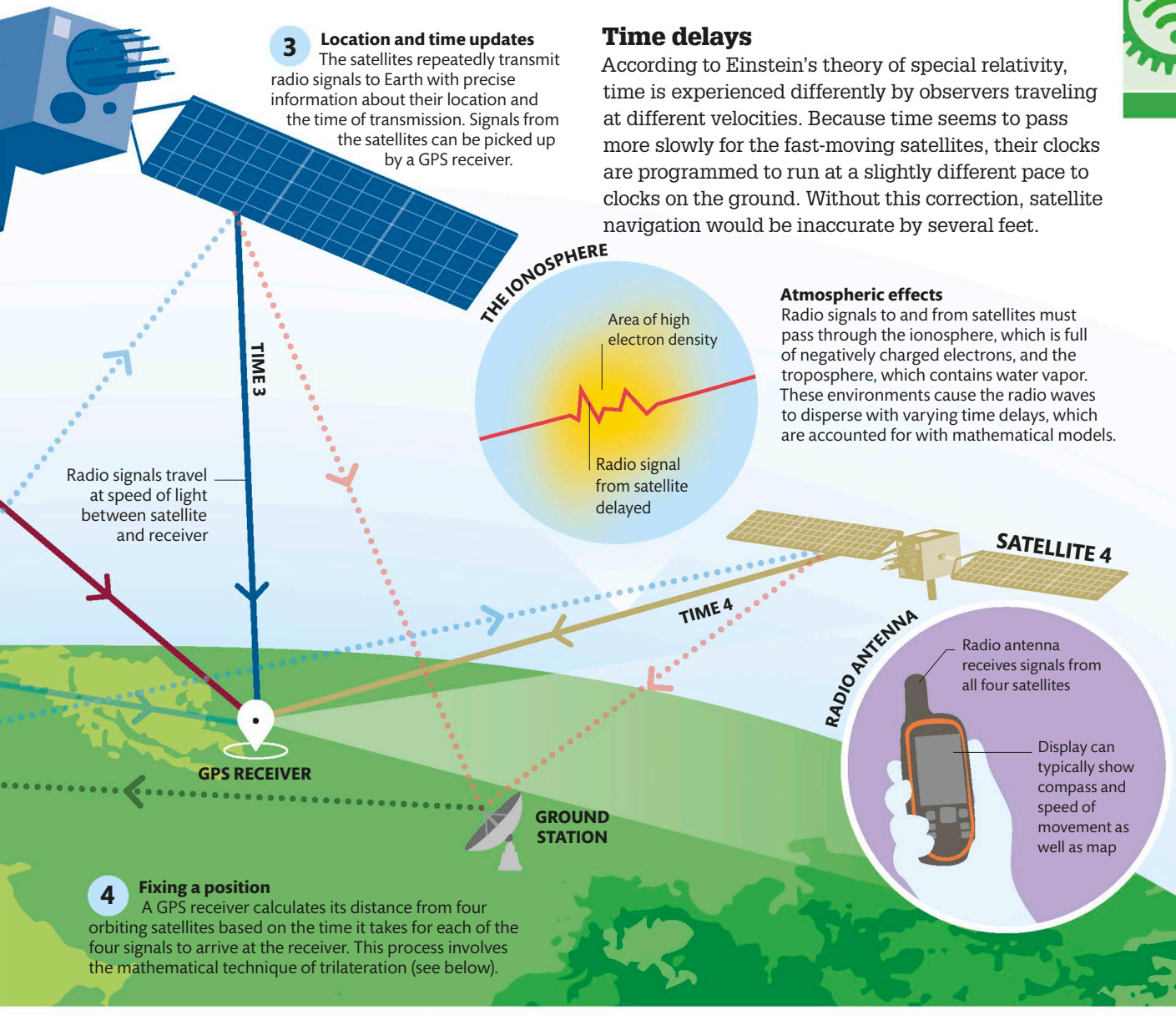
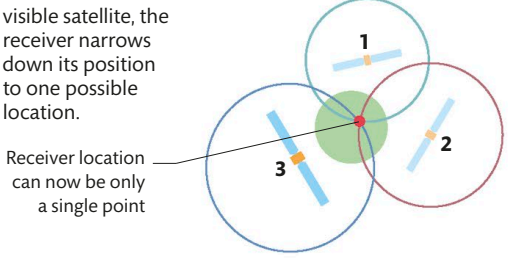
Time delays

According to Einstein's theory of special relativity, time is experienced differently by observers traveling at different velocities. Because time seems to pass more slowly for the fast-moving satellites, their clocks are programmed to run at a slightly different pace to clocks on the ground. Without this correction, satellite navigation would be inaccurate by several feet.



Satellite 3
As it calculates its distance from a third visible satellite, the receiver narrows down its position to one possible location.



Satellite 4
This satellite is used to correct the inaccurate position indicated by the receiver, because the clock built in to the receiver is not perfectly synchronized to the satellite clocks (see above).

